



Computer Vision and AI tools for automated pollinator monitoring

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General motivation



- **Declining trends in the diversity and abundance of pollinators** (especially insects) suggest future risks in meeting increasing global food demands (Potts et al, 2010; Cameron et al, 2011).
- **Managed pollinators** (e.g. honey bees) are main contributors of worldwide agricultural production (Potts et al, 2010)
- **Wild pollinators** are also critically important, as they increase agricultural yield and quality of fruit crops (Garraat et al, 2014).

Potential contributions of CV and AI

Automated pollinator monitoring

- For scaling up data collection
 - ▣ Longer periods, more sites, continuous autonomous monitoring
- For finer-grained data
 - ▣ Species abundance
 - ▣ Behavior: flower handling time, foraging patterns, foraging preferences, adaptation, individuality



4

Applicative overview

A journey in computer vision for pollinator monitoring



BigDBee Project (2016-2020)

- Large-scale video monitoring of honey bees at the ramp



DeepPollinator project (2020-2025)

- Monitoring in the field:
honey bees and wild pollinators



CyIndiBee Project (2023-)

- Cyberinfrastructure for individual bee monitoring:
ReID models, interfaces, real-time





BigDBee Project (2026-2020)

Large-scale video monitoring at the ramp



A multi-disciplinary project

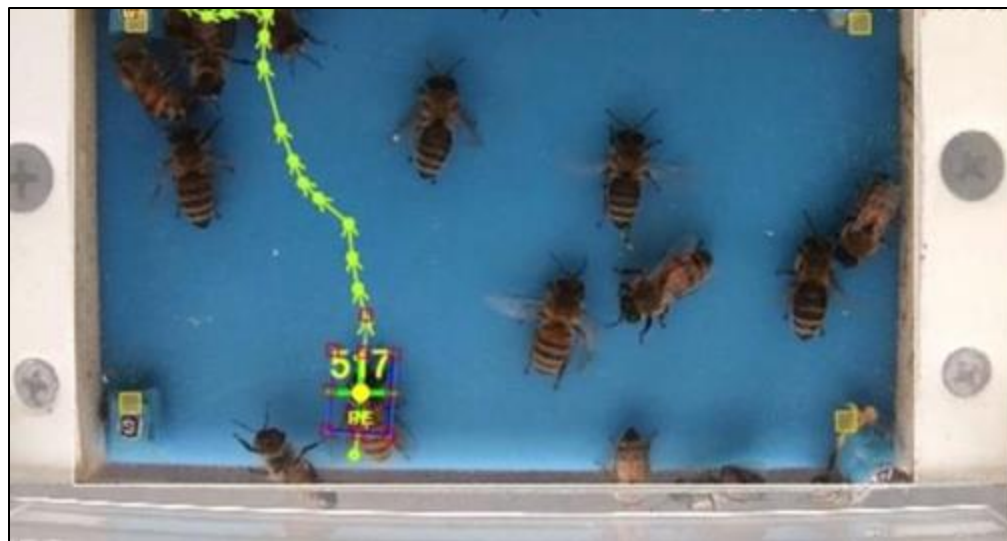
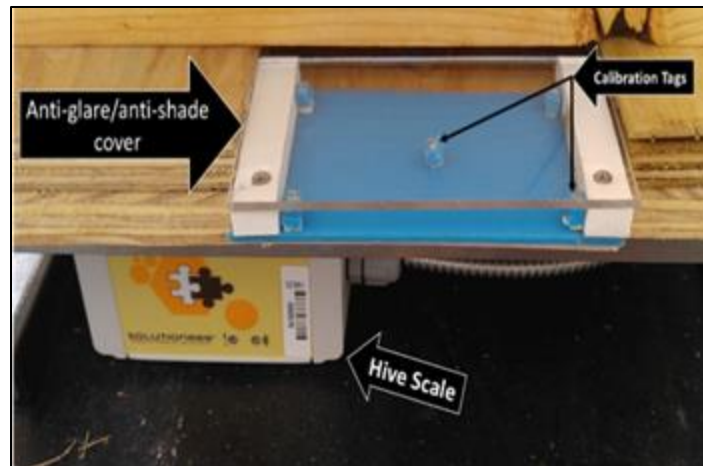
PI: Rémi Mégret (CS), Co-PI: Edgar Acuña (Stat)

PI: José Agosto (Biol), Co-PI: Tugrul Giray (Biol)

A



B



Honeybee behavior data collection

- Current methods are **labor intensive**



Observation
ramp



Number tags



Manual
annotation

- or look at only **part of the behavior** (entrance/exit, sensors)



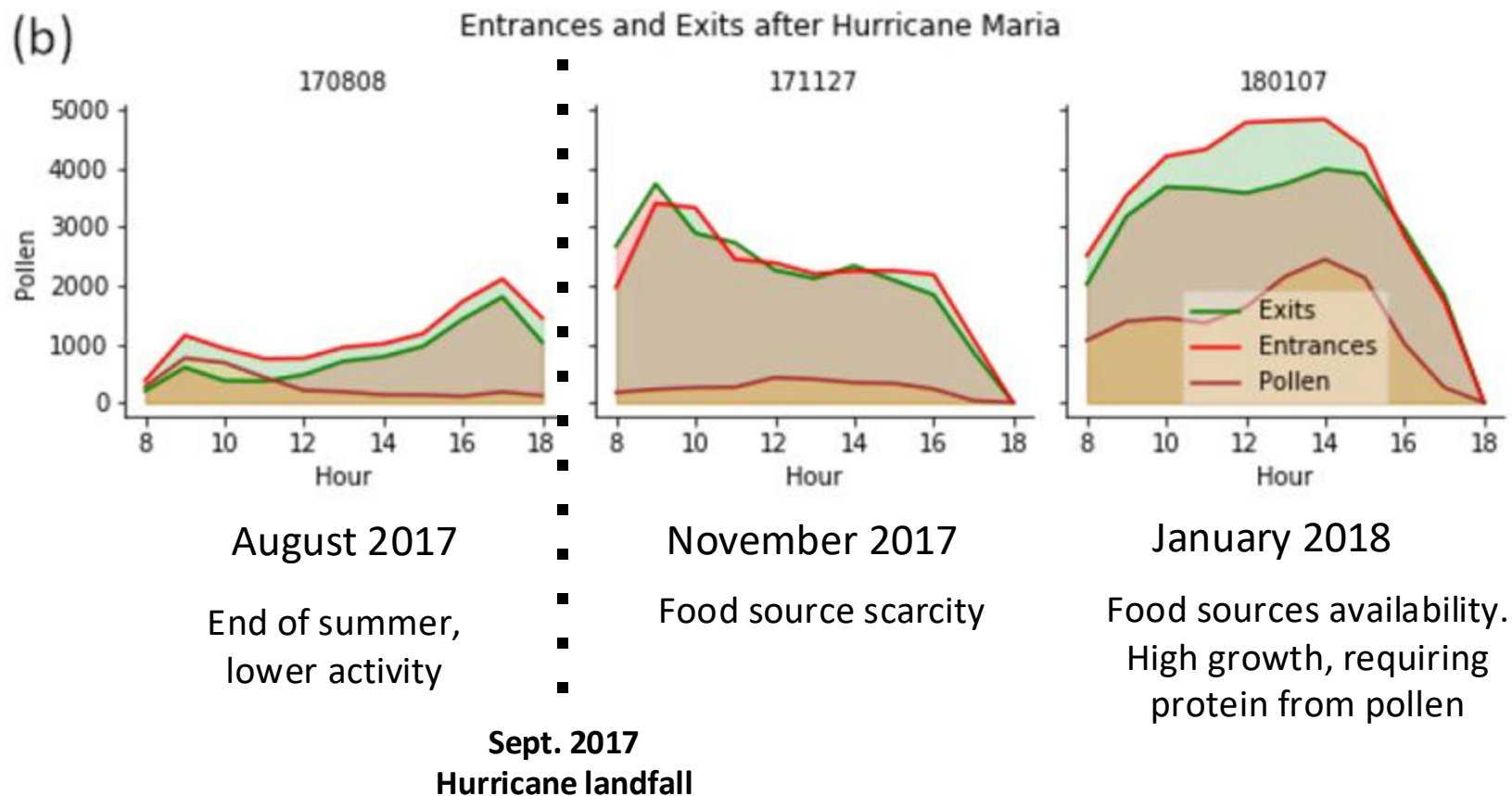
RFID



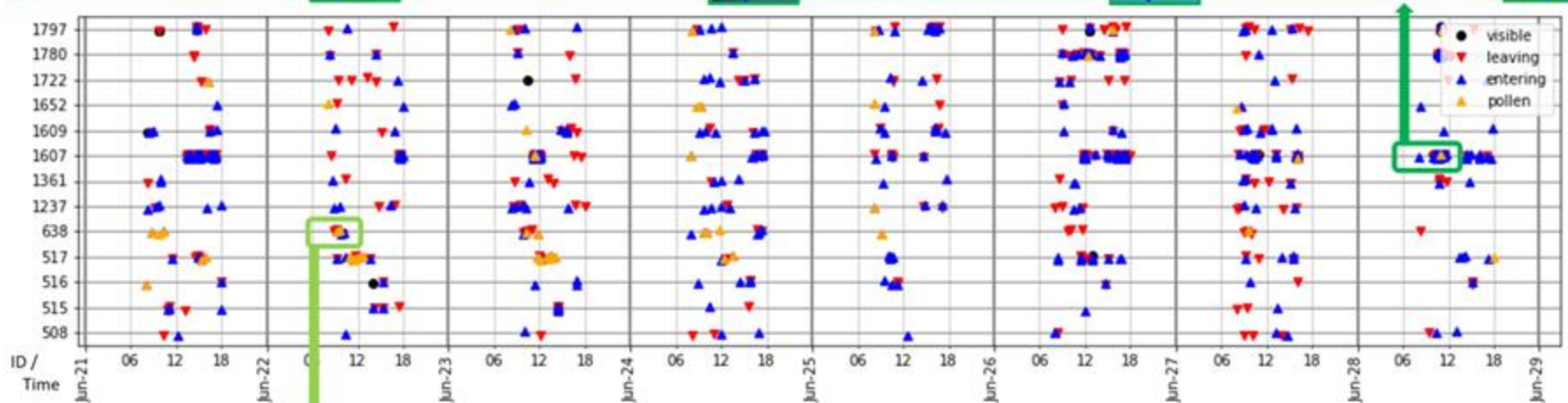
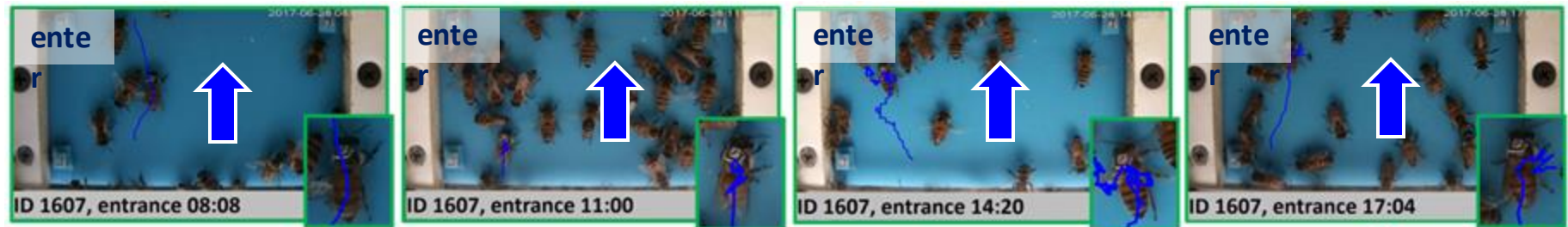
Optical counters

Colonywide behavior

e.g. Effect of hurricane Maria



Individual behavior



▶ Video Info

▶ Annotations

▶ I/O

▶ Features

▼ Zoom

☒ Update
 ☐ Overlay
 ☒ Box
 ☐ Tag

Z+

Z=

Z-

Export Image

▶ Alternate Tags

▼ Parts labeling

tail

+

-

Edit...

Preset...

● head

● thorax

● abdomen

● antL

● antR

● pollenL

● pollenR

● legL

● legR

Computer Vision and AI for automated pollinator monitoring

Monitoring in the field

PI: Rémi Mégret (CS), Co-PIs: Carlos Corrada (CS), José Agosto (Biol), Tugrul Giray (Biol), Elvia Melendez-Ackerman (Ecol)

Deep Learning for the analysis of insect pollinator behavior in the field.

Automatic insect detection, identification, and behavior classification from videos



Flower-Patch assay automation



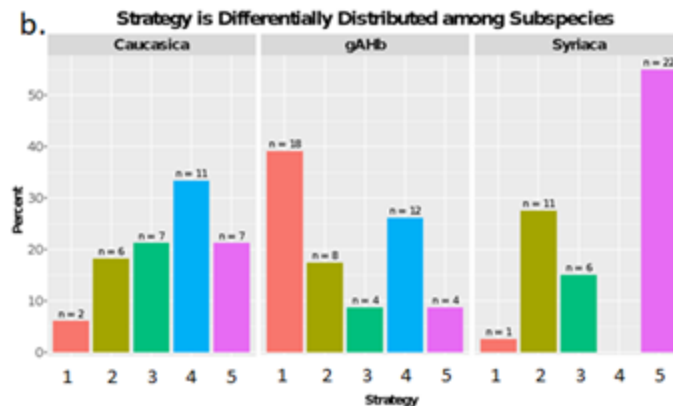
Monitoring of flower visits on mango tree



Visual monitoring of behavior is very low-throughput and prone to human error

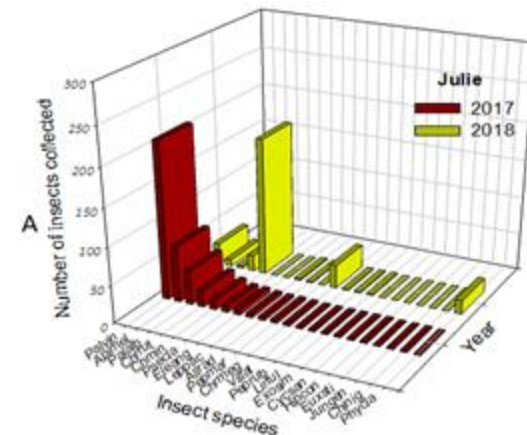
Motivation and impact

Managed pollinator (Honey bee)



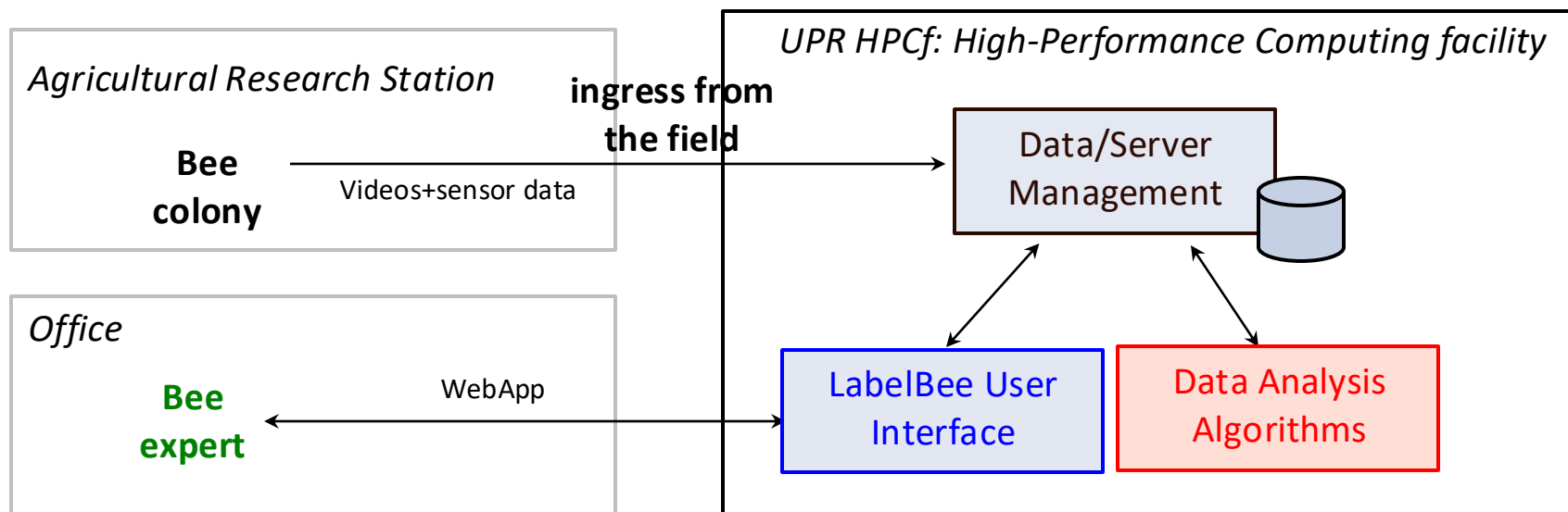
- Insights into decision-making and foraging strategies
- Breeding programs based on pollination behavior

Wild pollinators



- Quantification of wild pollinators abundance, flower visits, and correlation with yield

The need for cyber-infrastructure



Typical video sizes: 3-6GB/h ? 1-2TB/month of daytime/colony
Needs for annotation for model training, visualization, scalable computing...

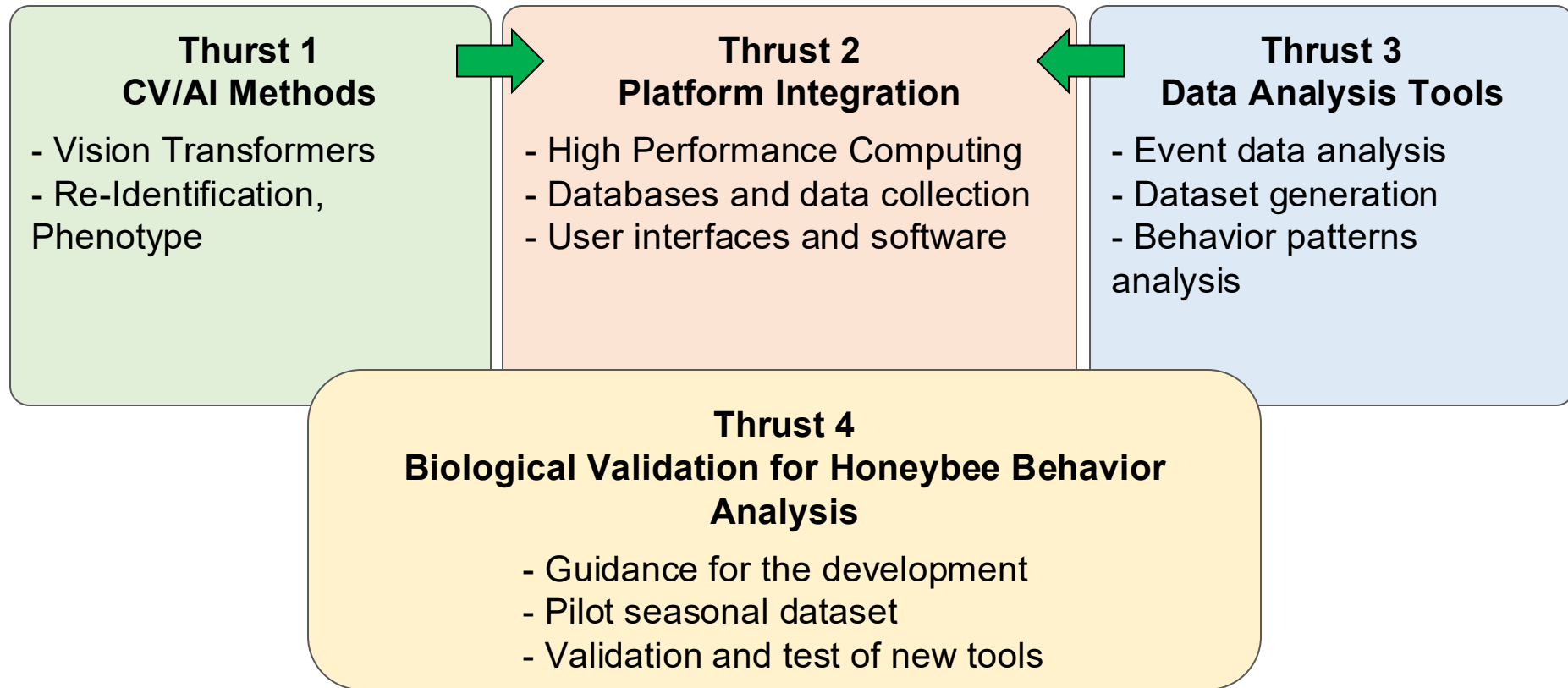


CyIndiBee Project (2023-)

Cyberinfrastructure for individual bee monitoring



PI: Rémi Mégret (CS), Co-PIs: Carlos Corrada (CS), Marie Lluberes (CS), José Agosto (Biol), Tugrul Giray (Biol)



Application to Bee Hunting, outreach

Map wild colonies by triangulation from azimuth of returning bees. Needs individual identification.

10th grade convergent research activities
(Collaboration CyIndiBee and AC3 projects)

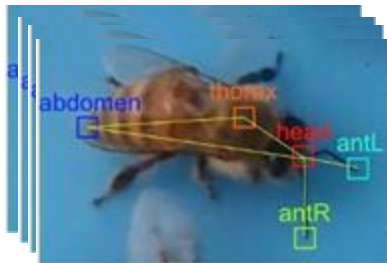


The CV and AI tools

- Offline monitoring
- Real-time monitoring
- Individual Re-ID

Offline monitoring

Pose Detection using Deep Learning



Training dataset

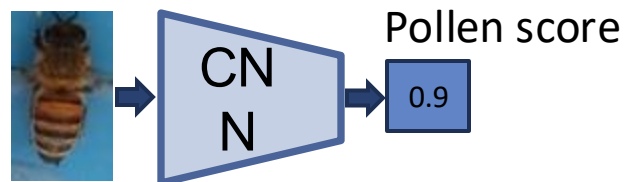


Automatic pose detection

Ivan F. Rodríguez, Rémi Mégret, Roian Egnor, Kristin Branson, José L. Agosto, Tugrul Giray, Edgar Acuña. "Multiple Insect and Animal Tracking in Video using Part Affinity Fields", Visual observation and analysis of Vertebrate And Insect Behavior (VAIB), International Conference on Pattern Recognition (ICPR), Beijing, China, 20-24 August 2018.

Pollen detection

- From classification
[Rodriguez 2018, 2022]

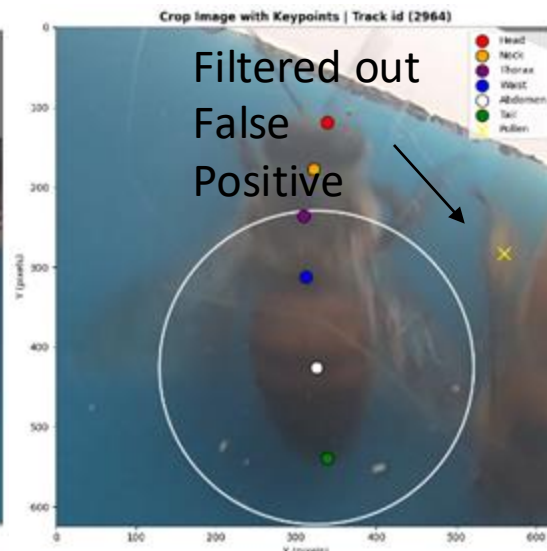
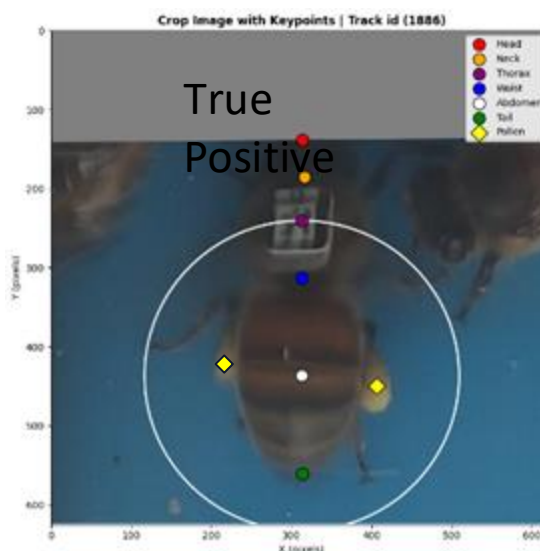
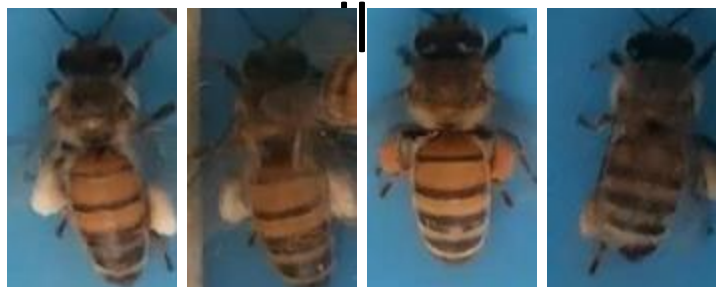


- To detection for improved granularity



Non-

Poll



Work of Pablo Campos

Ivan F. Rodriguez, Rémi Mégret, Edgar Acuña, Jose L. Agosto, Tugrul Giray. "Recognition of Pollen-bearing Bees from Video using Convolutional Neural Network", IEEE Winter Conference on Applications of Computer Vision, 2018.

Rodriguez IF, Chan J, Alvarez Rios M, Branson K, Agosto-Rivera JL, Giray T and Mégret R (2022) Automated Video Monitoring of Unmarked and Marked Honey Bees at the Hive Entrance. Front. Comput. Sci. 3:769338. doi: 10.3389/fcomp.2021.769338

Pollen characterization

Towards automatic characterization
of color, size

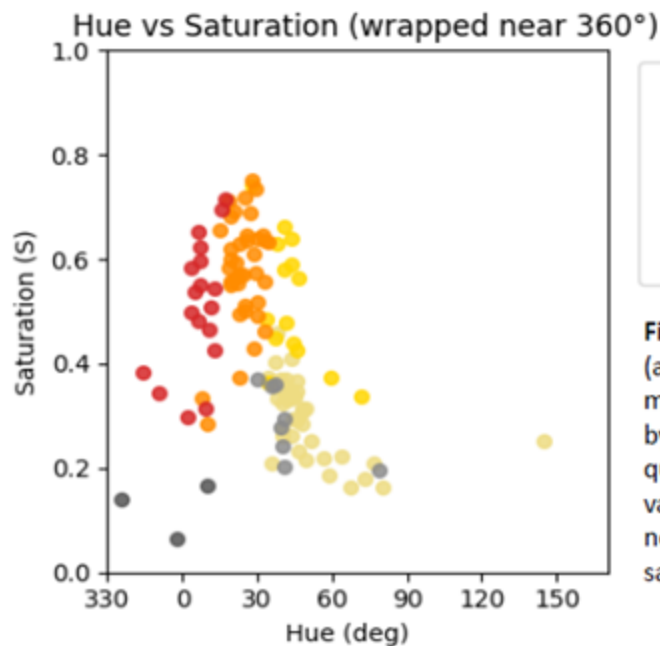
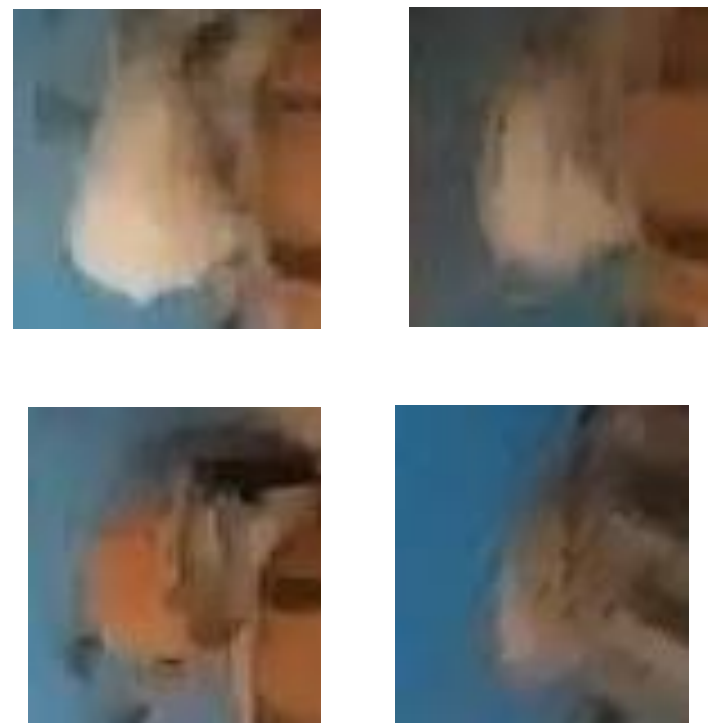


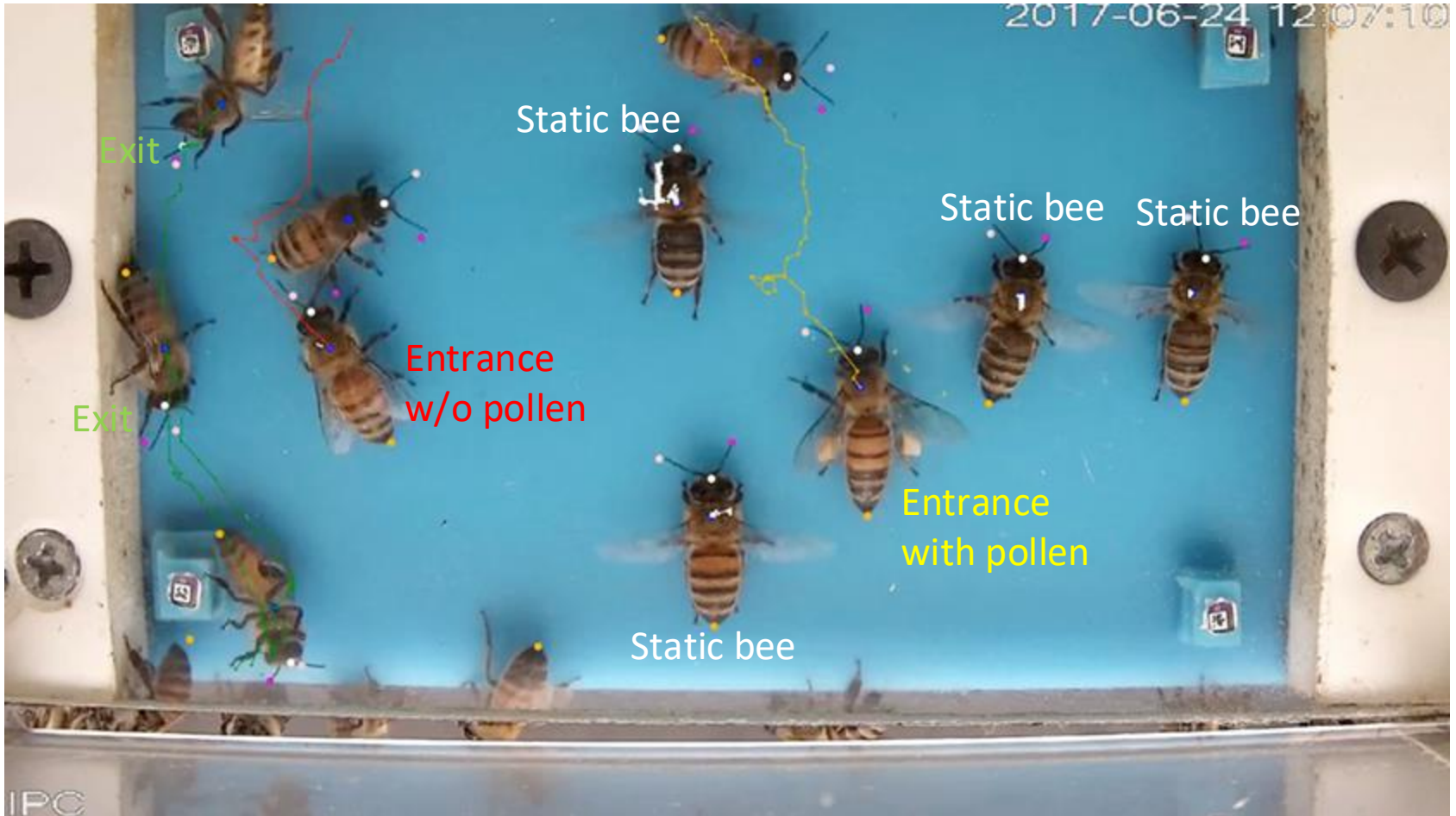
Figure 2: Polar check of path colors (angle = hue, radius = saturation, marker size = value). Points are colored by annotated class. The Plot provides a quick QC view to verify that human-validated patches cluster by neighboring tones and to spot low-saturation or mixed cases.



work of Sebastian Jansasoy

Jansasoy Molina S., Meyers L., Rodríguez J., Mégret R. Predicting Pollen Color in Bees with DINOv3: Patch-Level Embeddings and HSV/LAB Integration. SIDIM 2026

Track-level behavior classification



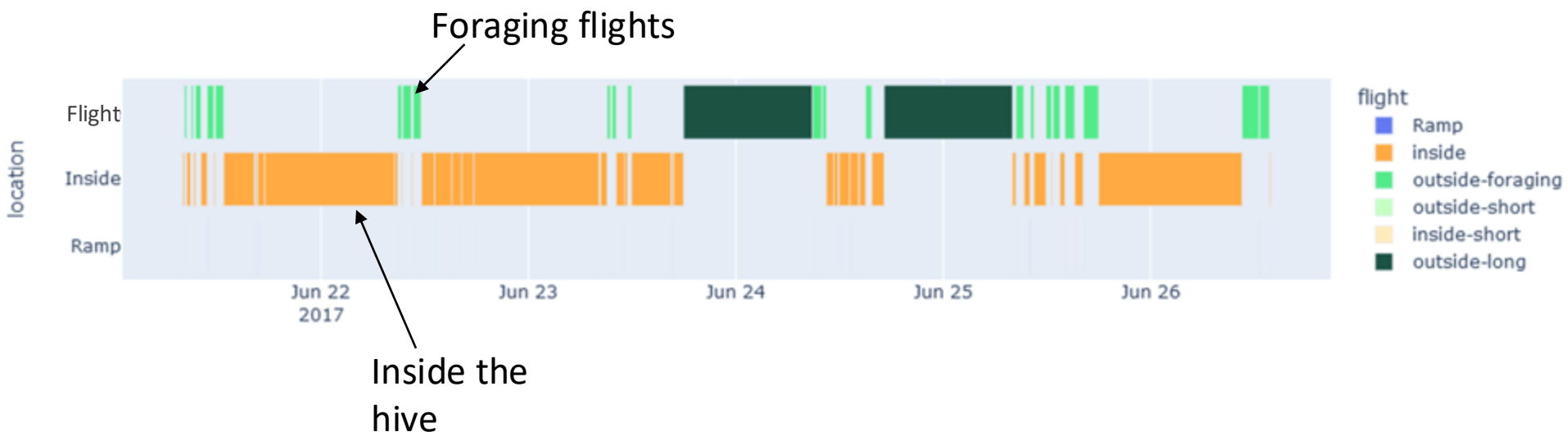
Flights reconstruction

MS thesis
Andrea Nieves
Collaboration
Dr. Marie
Lluberes

□ Pairs

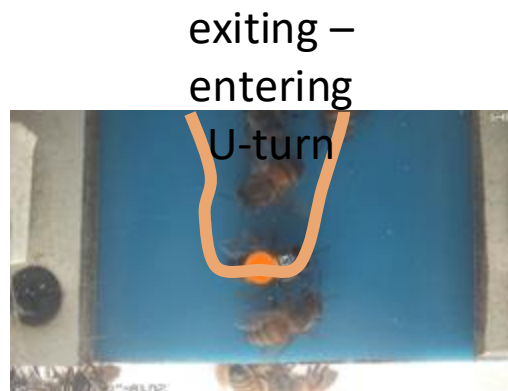
□ exit□entrance: flight trip

□ entrance□exit: inside

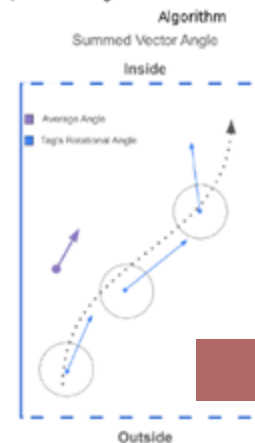
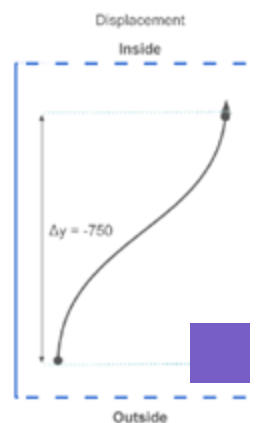
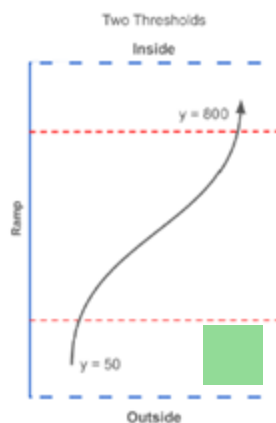
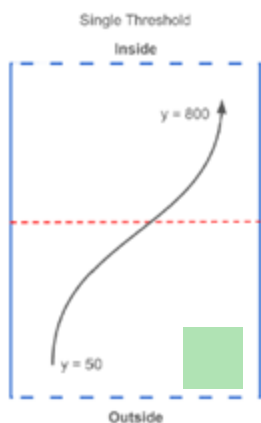
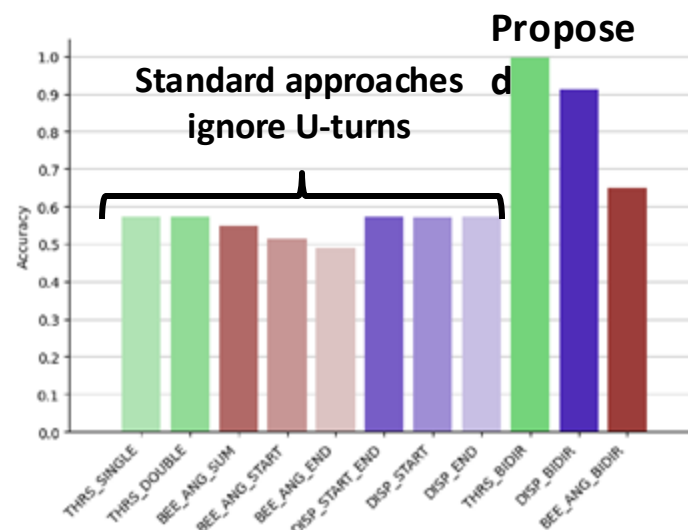
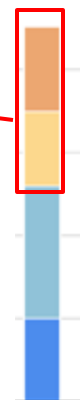


Entrance/Exit is not enough

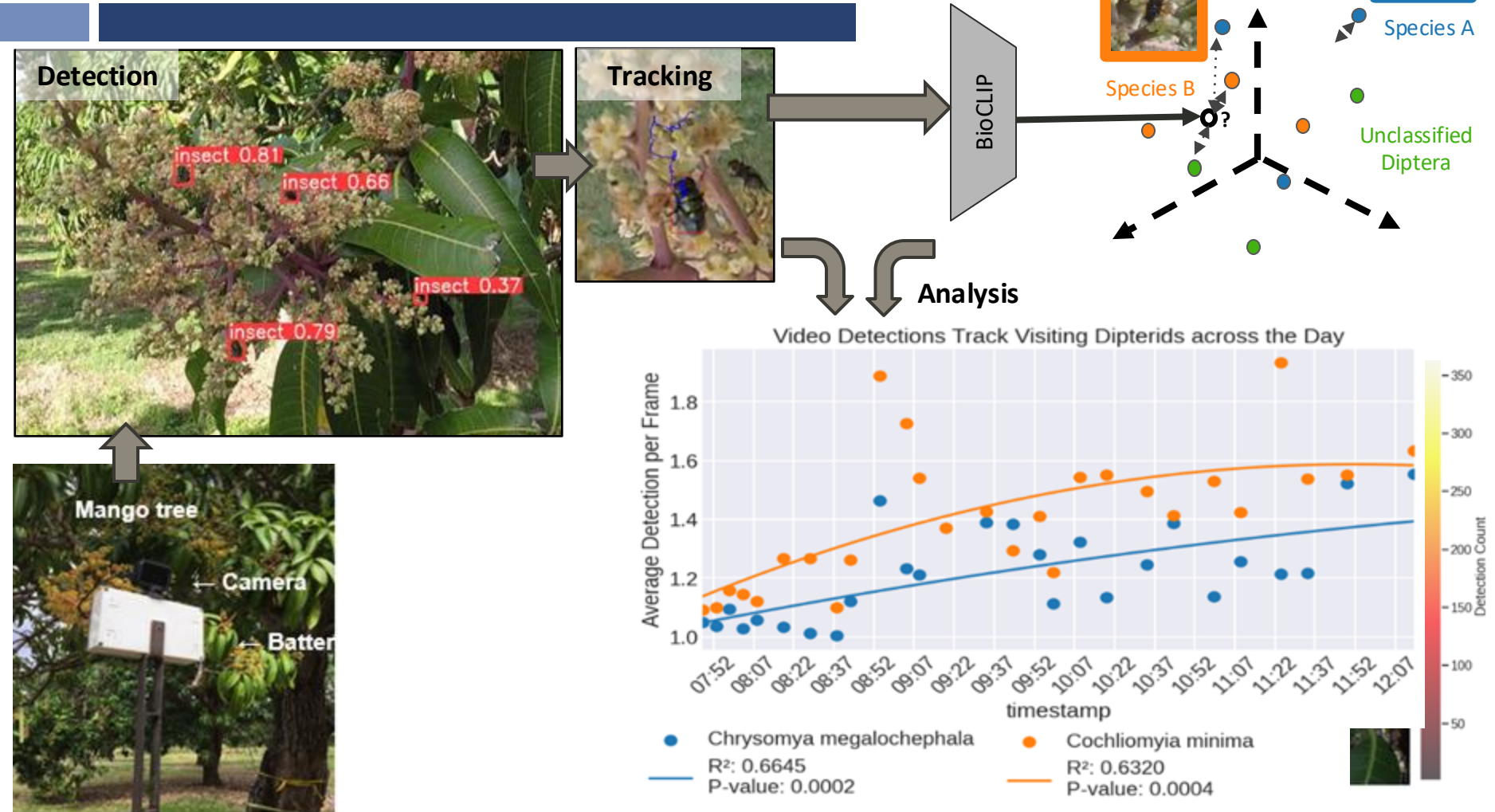
MS thesis
Andrea Nieves
Collaboration
Dr. Marie
Lluberes



more than
35% of events
are U-turns.



Offline analysis: Wild pollinator setting



L. Meyers, J. Rodriguez-Cordero, K. Garcia, I. Cabrera, El Melendez-Ackerman, R. Megret. Video Analysis and Machine Learning Tools Enable Fine-grained Tracking of Pollinators of Mango *Mangifera indica* in Puerto Rico. ESA Meeting, Aug 2025.

Species recognition of mango pollinators: the importance of resolution



1024 x 720 0.7Mpix GoPro



2816 x 2816 8Mpix Basler Cam

L. Meyers, J. Rodriguez-Cordero, K. Garcia, I. Cabrera, El Melendez-Ackerman, R. Megret. Video Analysis and Machine Learning Tools Enable Fine-grained Tracking of Pollinators of Mango *Mangifera indica* in Puerto Rico. ESA Meeting, Aug 2025.

Real-time monitoring

Flowerpatch assays setup



Traditional
setup



Artificial
flowers



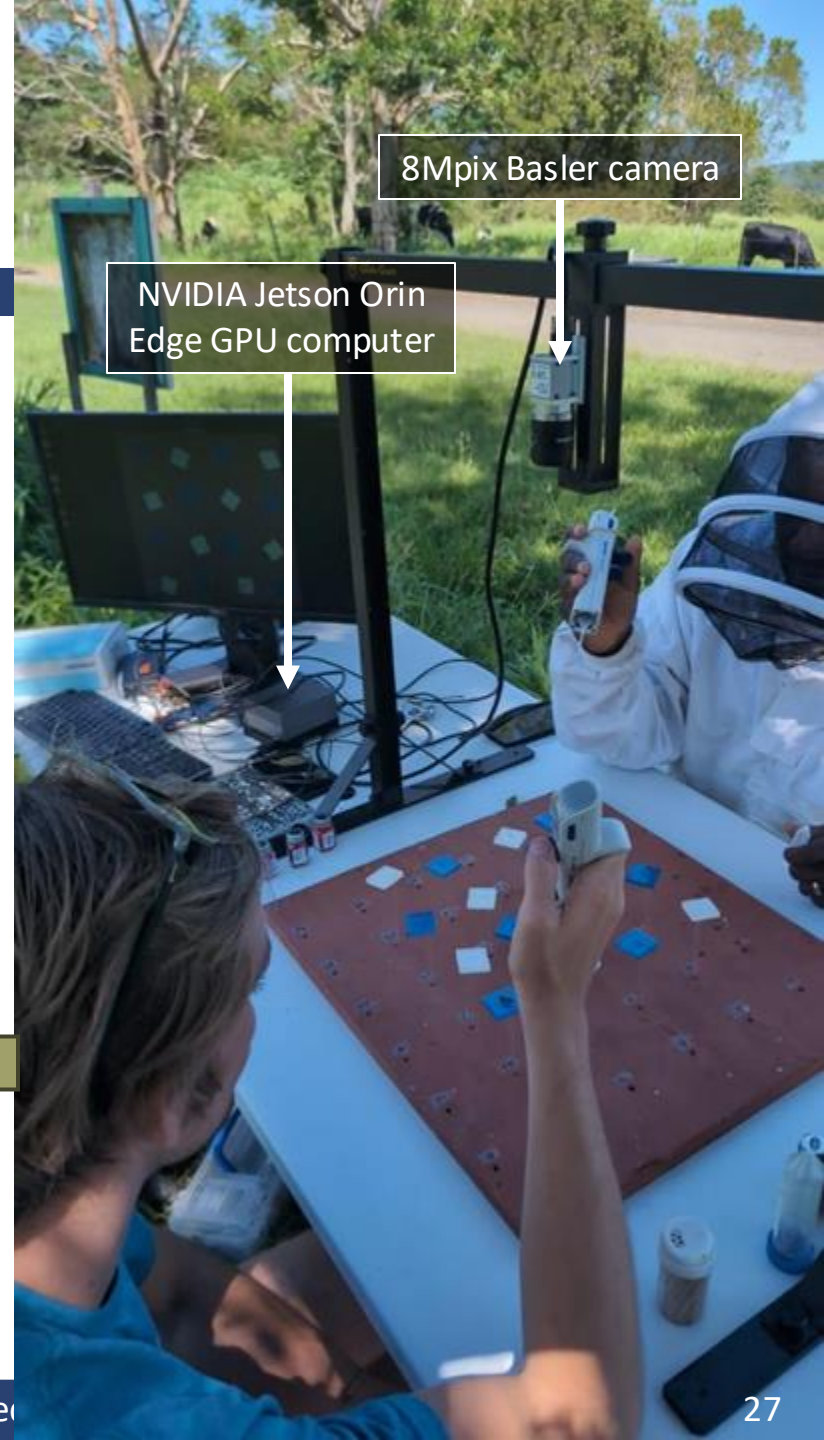
Paint
markings
for ReID



Pose

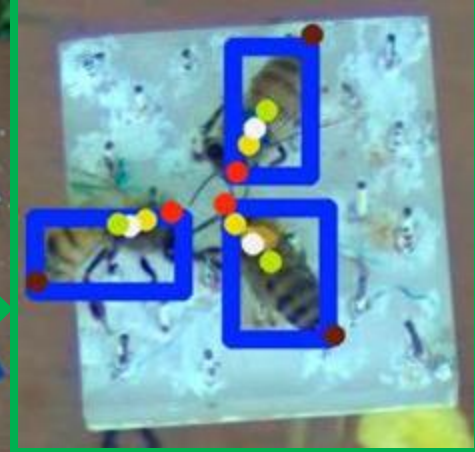
detection

Computer vision and AI for automated

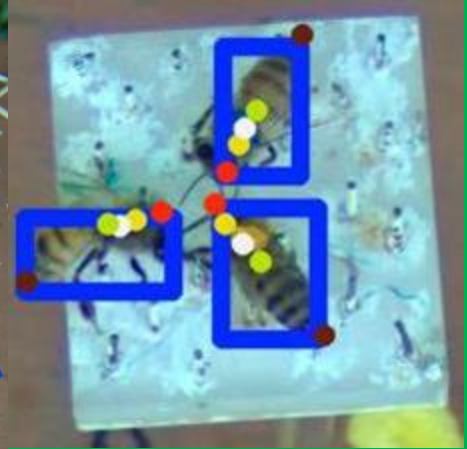


8Mpix Basler camera

NVIDIA Jetson Orin
Edge GPU computer



**1 visit =
head in the well
for more than 1s**

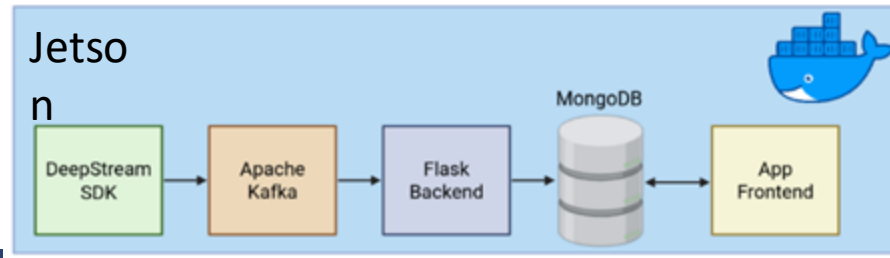


Real-time detection
in the field:

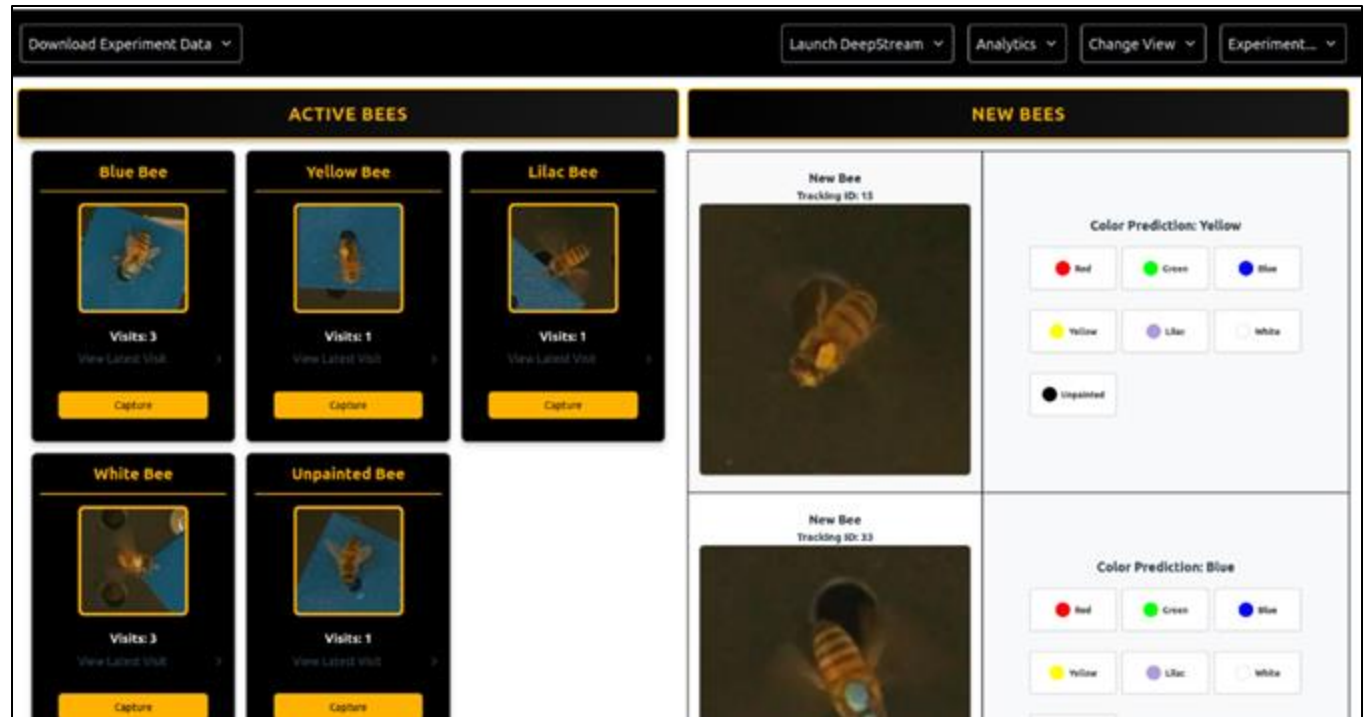
- Orin AGX:
- 8MP camera, 20fps
- YOLOv8n pose

[VIDEO \[Google drive\]](#)

Real-time system + interface



Key constraint: real-time visit count by ID to capture each bee after 50 visits



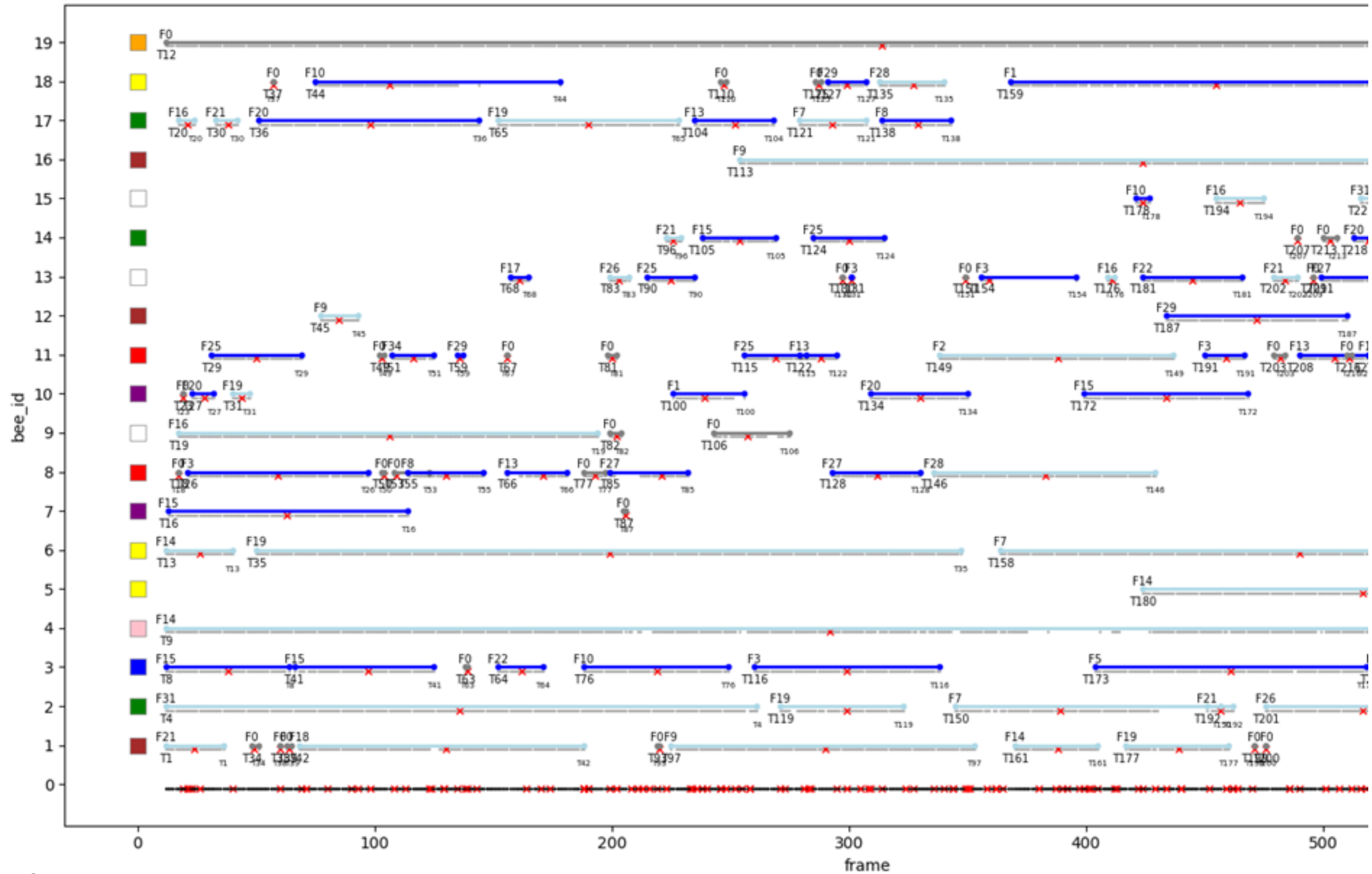
Real-time visit detection and ID validation

Assay setup
calibration

Contributors:
Nestor Martinez
Josue Rodriguez
Gabriel Santiago
Adriana Hernandez
Eduardo Figueroa

Josué A. Rodríguez-Cordero, et al.. " A Real-Time Edge Computing System for Monitoring Bees at Flowers". ISICN 2024. [10.1007/978-3-031-67447-1_27](https://doi.org/10.1007/978-3-031-67447-1_27)
Néstor A. Martínez Rivera, Josué A. Rodríguez Cordero, Remi Megret Laboye, A Real-Time, Interactive Web Application for Automated [Bee]havioral Analysis of Honeybees, SIDIM 2026

Visit data ☐ behavioral statistics per ID



Individual re-identification

How to Re-ID a small insect *individually* ?

Robust,
more invasive

Less invasive,
requires CV
models



apriltag



number
tag



painted
color
code

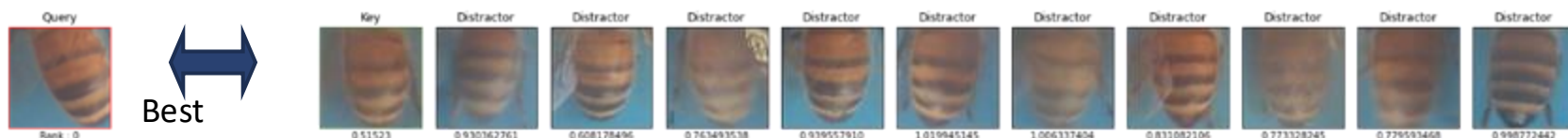


painted blob

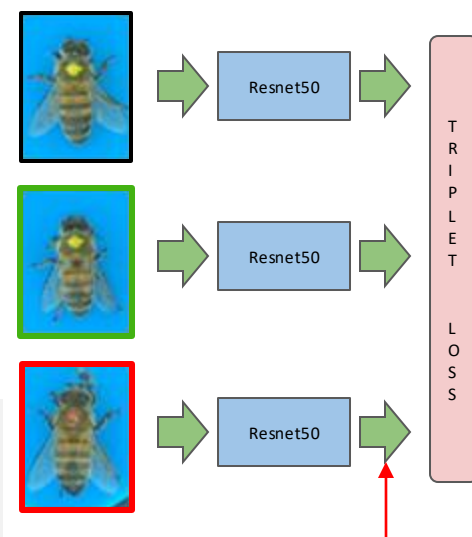
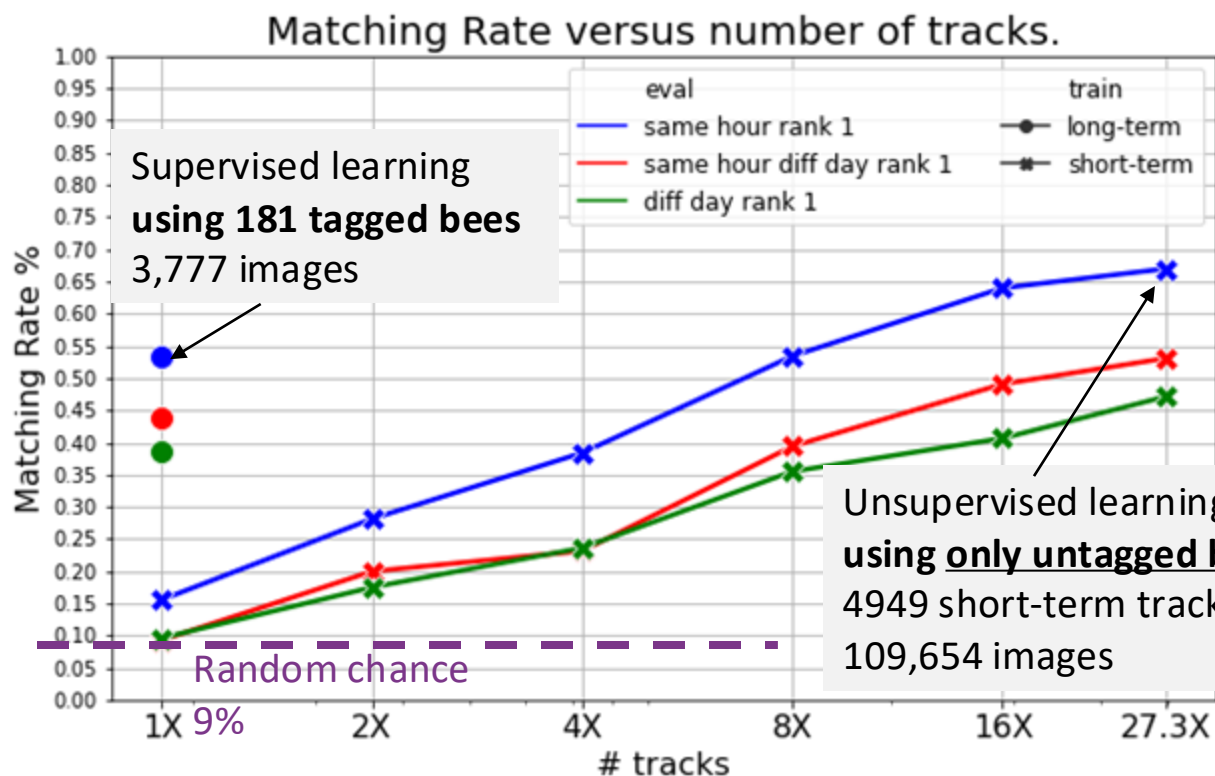


no
marking

Contrastive Learning for unmarked ReID



(Gallery of 1 pos, 10 neg)



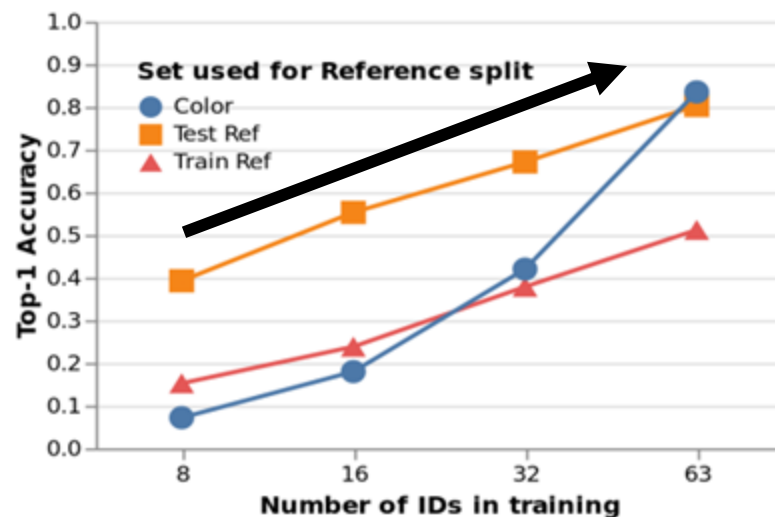
Still huge amounts of training data

Latent space of identity

J. Chan et al. Honeybee re-identification in video: new datasets and impact of self-supervision. VISAPP 2022

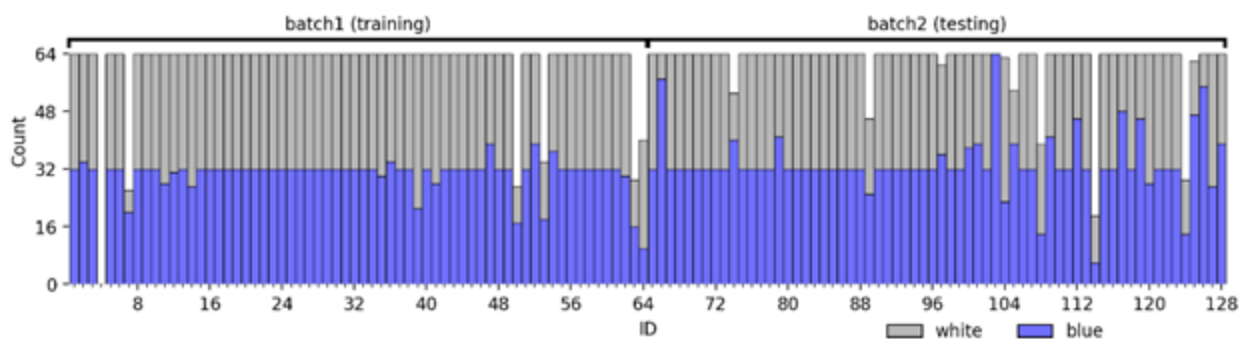
MS thesis
Jeffrey Chan

Lighweight Re-ID with paint markings



**Best performance
requires large data
collection effort!**

Dataset of 64 train IDs
+ 64 test IDs with
matching color codes



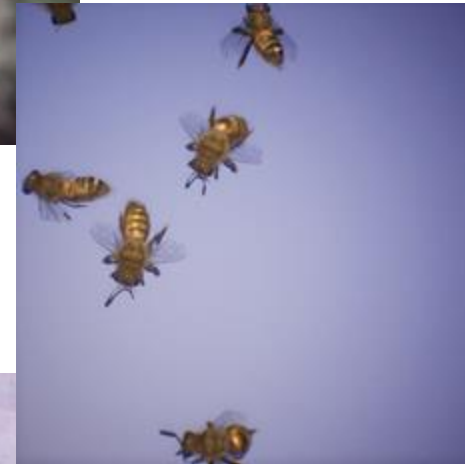
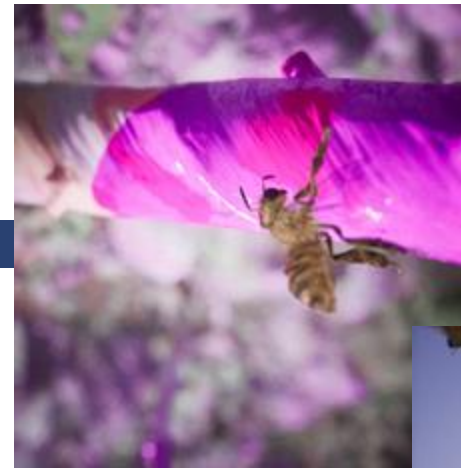
Synthetic datasets

- Adaptation of replicAnt [Plum 2023] to honeybee can generate large training datasets with custom markings
- Synthetic data was shown to complement natural data in low data regime [Matos 2026] but not replace



C. Matos, D. Flores, R. Megret. A Parametric 3D Bee Model for Scalable Synthetic Data Generation in Animal Behavior Studies. VISAPP 2026.

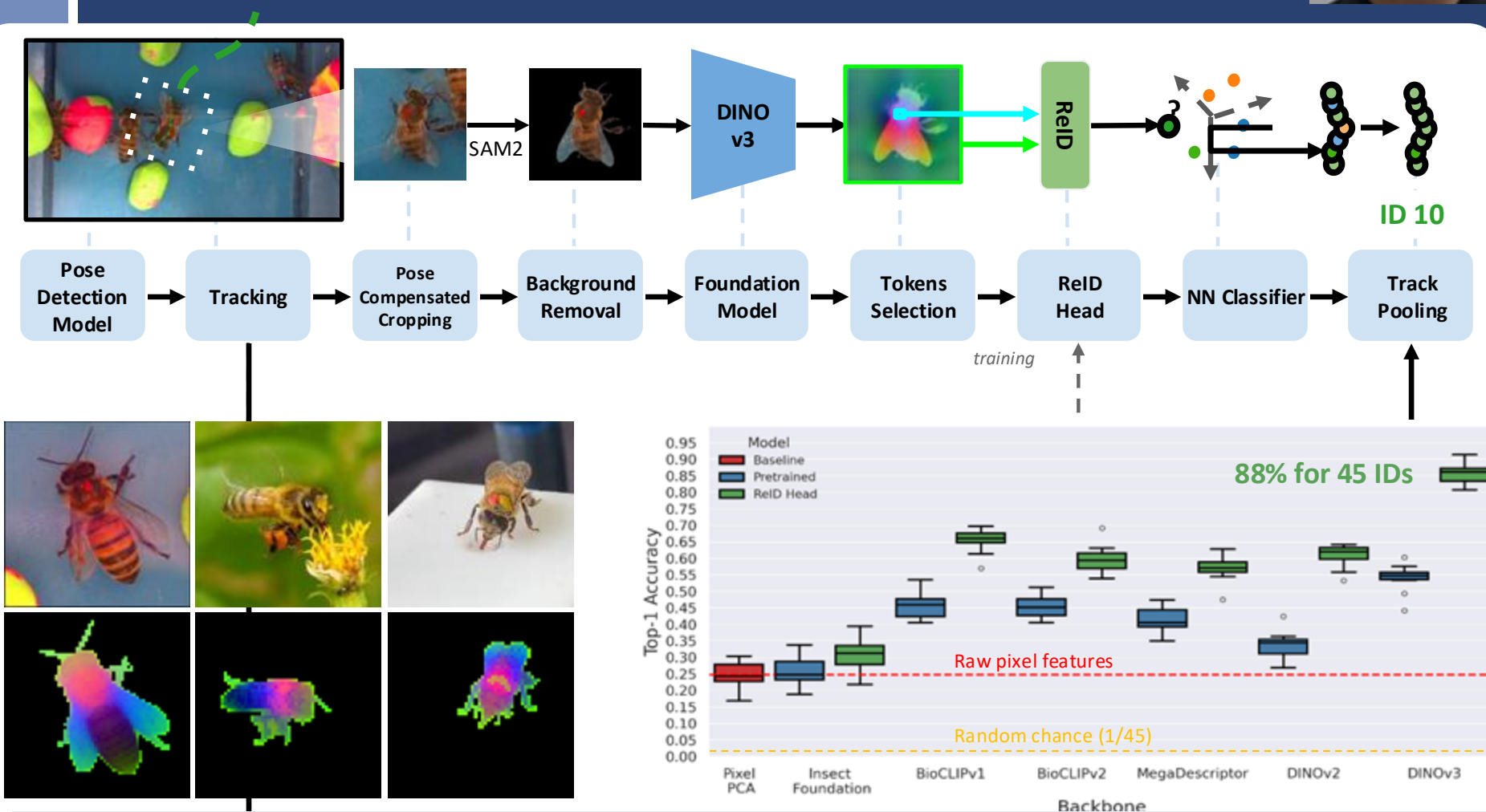
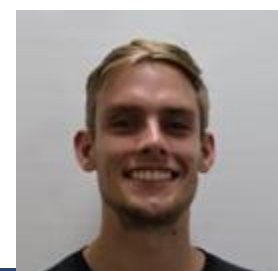
Plum, F.; et al. replicAnt: A Pipeline for Generating Annotated Images of Animals in Complex Environments Using Unreal Engine. *Nat Commun* **2023**, 14 (1), 7195. <https://doi.org/10.1038/s41467-023-42898-9>.



MS thesis Christian Matos
Collaboration Dr David Flores

Leveraging Foundational Models for ReID

(see poster #4 of Luke Meyers, poster session 1 this Sunday)



L. Meyers, J. Rodriguez-Cordero, R. Megret. One-Shot Fine-Grained Re-Identification of Paint Marked Honey Bees using Vision Foundation Models, WACV 2026

Conclusion

- Deep-Learning and Foundational models: a revolution in our capacity to automate pollinator behavior monitoring
 - ▣ Accurate detection and pose estimation, classification, re-id
 - ▣ Real-time/low-latency in the field possible with powerful edge GPU
- Computational Infrastructure challenges for scalable deployment
 - ▣ Handling of massive TB datasets
 - ▣ Need for integrated visualization, analysis and annotation tools
- Open questions
 - ▣ Less constrained setups (outdoor, crowd, open environment...)
 - ▣ Scalability/robustness/cost for long-term automatized monitoring?
 - ▣ Foundational models suitable for fine-grained agricultural data analysis

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Hector Carrion
Joel Gonzalez
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Pablo Campos
Pablo Torres-Arroyo
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Skyler Williams

...

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